# Zengtao Liang

Boca Raton, Florida | (917) 470-8026 | No Sponsorship Needed | https://zeitan0.github.io/my-portfolio/

## SKILLS AND INTERESTS

- *Computer Skills:* Java, JavaScript, Python, C++, Rust, SQL, R, HTML, React.js, Node.js, MongoDB, Unity, Blender
- *Language Skills:* bilingual in Chinese (Mandarin) and English
- *Certifications:*
  - *Microsoft Technology Associate: Introduction to Java*
  - *Adobe Certified Professional: Adobe Photoshop 2018*
  - *Microsoft Office Specialist: Excel, Word, and PowerPoint*

## EDUCATION

**University of Florida,** Gainesville, Florida — Expected Graduation: May 2027

*BS in Computer Science* — *GPA:3.97/4.00*

*Relevant Coursework: Operating Systems, Computer Structure and Algorithms, Intro to Software Engineering, Info and Database Systems, Natural Language Processing*

*Awards:* Dean's List

## EXPERIENCE

**Coding Tutor,** Gainesville, FL — June 2023 – Present

- Provide personalized one-on-one tutoring in multiple programming languages, including C++, Java, and Rust, to students ranging from beginners to advanced learners.
- Helped over 20 students improve coding proficiency, with many achieving higher grades, successful project completions, or programming competition placements.

**Chinese Learning Assistant,** Boca Raton, FL — June 2021 – June 2023

- Assisted children of various ages in learning Mandarin Chinese, developing skills in speaking, listening, reading, and writing.
- Helped students achieve measurable improvements in language proficiency, with several demonstrating enhanced confidence in conversational Mandarin.

## PROJECTS

**FUSE WAD File System,** C++ — August 2025

- Designed and implemented a user-space file system using FUSE, enabling mounting and navigation of WAD archive files as a hierarchical directory structure.
- Built a custom WAD parser to read and interpret on-disk descriptors, directory entries, and lump metadata
- Diagnosed and resolved low-level I/O and descriptor offset bugs that caused corrupted reads and improper metadata mapping
- Implemented directory traversal, file extraction, and writable operations with correct inode metadata persistence

**Task Management Web App,** MERN Stack — May 2025

- Developed a full-stack task management application using MongoDB, Express.js, React.js, and Node.js (MERN).
- Designed and implemented RESTful APIs for task CRUD operations and deadline tracking.
- Developed a responsive React frontend with dynamic state management and real-time UI updates
- Integrated MongoDB for persistent data storage with structured schema design

**Lua Interpreter,** Rust — May 2024

- Implemented core components of a Lua interpreter in Rust, including lexical analysis and tokenization
- Designed foundational parsing infrastructure to support future AST construction and execution
- Laid groundwork for executing Lua scripts efficiently, with robust error handling and modular design for extension.